Provigil AI AETHER SIEM & XDR Platform

Complete Migrated Rust Feature Catalog + Next-Generation Innovative Roadmap

Version 2.1 | 100% Implemented & Verified in Rust (37 Crates) | Next-Gen AI-Native Capabilities Added

Attribute	Specification
Platform	backend-rust + provigil-common (37 Modular Cargo Crates)
Status	Production Core 100% Implemented | Next-Gen Roadmap Integrated
Architecture	High-Throughput Rust (Tokio Async) + ClickHouse + Angular SOC Console
API Daemon	Active on Port 8088 (REST + WebSocket)
Build Status	Clean Workspace Build | Zero Warnings | 37/37 Crates Verified

EXECUTIVE NOTICE: *This document is the definitive catalog of all features fully migrated into production-ready Rust, plus the strategic Next-Generation Innovative Feature Roadmap required to elevate AETHER SIEM into the top tier of autonomous, AI-native security operations platforms.*

PART A — PRODUCTION FEATURES (100% Implemented in Rust)

1. Autonomous AI Parser Synthesizer (siem-parser-gen)

Eliminates manual regex authoring for unknown log formats via a two-tier architecture: nanosecond hot-path for known formats and asynchronous AI synthesis for new ones.

- **CityHash-64 Hot Path:** Strips dynamic tokens (timestamps, IPs, PIDs), computes 64-bit structural fingerprint, O(1) lookup at ~0.8 µs latency. Zero LLM cost for recurring formats.
- **Sample Buffer Threshold:** Unknown fingerprints accumulate; synthesis triggers at exactly 20 samples.
- **Dual-Engine Synthesis:** Groq Llama-3 / OpenAI for cloud mode; offline heuristic regex inducer for air-gapped deployments.
- **Sandbox Verification:** Candidate regexes tested against all 20 samples; ≥80% accuracy required before registry promotion.
- **Parser Studio:** Angular UI for schema inspection, hot testing, manual editing, and persistent storage to learned_parsers.json.

2. Detection & Correlation Engine (siem-engine + provigil-common)

Full Wazuh rule parity plus cross-source correlation and NDR corroboration.

- **1,435 Official Wazuh XML Detection Rules** compiled and evaluated in-memory with hierarchical inheritance (if_sid, if_group, if_matched_sid).
- **1,307 Multi-Protocol Decoders** (Syslog, Windows EventChannel, Apache, NGINX, SSH, Auditd, Cisco, Palo Alto, Fortinet, etc.).
- **10 OOTB Cross-Source Correlation Rules:** Brute Force, Account Compromise, Lateral Movement SMB, Data Exfiltration, Privilege Escalation, Impossible Travel, C2 Beacon, Malicious Macro, DNS DGA, Admin Backdoor — all MITRE-mapped.
- **NDR Corroboration Engine:** Cross-correlates Suricata/Zeek perimeter alerts with host telemetry; escalates to High/Critical.
- **Full MITRE ATT&CK; v14** tagging across tactics, techniques, and sub-techniques.

Rule	Name	Logic	MITRE
100101	Brute Force	≥5 failed auths in 60s	T1110
100102	Account Compromise	Fail then success in 10m	T1110+T1078

Rule	Name	Logic	MITRE
100103	Lateral Movement SMB	SMB to ≥3 hosts in 5m	T1021.002
100104	Data Exfiltration	>100MB outbound in 10m	T1048
100105	Privilege Escalation	User created + admin in 5m	T1136+T1078
100106	Impossible Travel	Distant geos in 1 hour	T1078
100107	C2 Beacon	Periodic ≤10% jitter	T1071+T1102
100108	Malicious Macro	Office spawns CMD/PS	T1059+T1204
100109	DNS DGA	High Shannon entropy	T1568.002
100110	Admin Backdoor	Create + logon in 5m	T1136+T1078

3. Live Vulnerability Detection (siem-vuln-detector)

- Thread-safe hot CVE feed sync via POST /api/v1/vulnerabilities/sync-feed (Arc<RwLock>).
- Continuous package inventory evaluation against CVE rules as syscollector data arrives.
- Official Wazuh Rule 23501 alerts with CVE ID, CVSS, package, version, remediation.
- Real-time WebSocket push to Angular SOC Vulnerabilities tab.

4. Security Configuration Assessment & CIS Benchmarks

8 native CIS checks on Windows with automated Rule 19001 alerting and real-time scoring via GET /api/v1/agents/:id/sca.

CIS Check	Mapping	Target
CIS 2.3.17.1	SOC 2 CC6.1	UAC EnableLUA
CIS 18.9.1	PCI DSS 5.2.1	Defender Real-Time Protection
CIS 18.2.1	NIST CM-7	SMBv1 Disabled (EternalBlue)
CIS 18.9.2	PCI DSS 2.2.4	RDP NLA Enforcement
CIS 9.1.1	CIS Controls	Windows Firewall Enabled
CIS 2.3.11.1	SOC 2 CC6.8	LSA Credential Guard (RunAsPPL)
CIS 18.5.1	NIST SC-8	LLMNR Multicast Disabled
CIS 18.9.30	PCI DSS 5.1	SmartScreen Enforcement

5. Deep Host Telemetry & System Inventory

- Windows: Recursive 64-bit + 32-bit Uninstall registry enumeration (DisplayName, Version, Publisher, InstallDate).
- Linux: Native /var/lib/dpkg/status parser for Debian/Ubuntu packages.
- Hardware, network (MAC/IPv4/IPv6), listening ports, services, user sessions.
- Normalizer converts OS-specific structures into standard syscollector format.

6. Real-Time File Integrity Monitoring (siem-syscheckd)

- Persistent SHA-256 baselines for hosts file, Startup folders, PowerShell profiles.
- NTFS Alternate Data Stream (ADS) detection for hidden execution streams.
- Rust notify crate for instant kernel filesystem events.
- Rules 550 (Added), 554 (Modified + hash diff, MITRE T1565.001), 553 (Deleted).

7. Rootkit & Kernel Anomaly Detection (siem-rootcheck)

- Hidden process detection via OS API vs kernel object list comparison.
- Trojan port scanning against known malware backdoor signatures.
- Hidden filesystem artifacts and unsigned driver detection.

8. Wazuh Agent Enrollment & Cryptography

- 192-bit key derivation: Key = MD5(RawKey) || MD5(MD5(Name) || MD5(ID))[0..15].
- siem-authd on port 1515: OSSEC enrollment handshake OSSEC K:'ID NAME IP KEY'.
- siem-remoted on port 1514: UDP/TCP Blowfish + AES-128/192/256 encrypted payloads.
- Thread-safe in-memory keystore with persistent client.keys synchronization.

9. SOAR Active Response & Threat Containment

- Windows: netsh advfirewall IP drop rules; Linux: iptables/nftables DROP.
- User account lockdown via net user /active:no.
- Non-blocking 600-second Tokio auto-reversion timer to prevent lockouts.
- siem-integratord: Slack, PagerDuty, Jira, VirusTotal webhooks for severity ≥10.

10. Multi-Format Syslog Forwarding (siem-csyslogd)

- CEF (Common Event Format) for ArcSight/Micro Focus.
- Newline-delimited JSON for Splunk HEC / Elastic Logstash.
- Splunk key/value (splunk-kv) and standard RFC 3164 BSD syslog.

11. Regulatory Compliance Evidence & Reporting

- PCI DSS v4.0.1, SOC 2 Type II, ISO/IEC 27001:2022, HIPAA Security Rule, NIST CSF 2.0 mappings.
- GET /api/v1/reports/summary for live compliance metrics and rule frequency.

12. High-Performance Columnar Storage & Resiliency

- ClickHouse tables: unified_alerts, siem_events, siem_alerts, xdr_incidents, ndr.threat_intel.
- 1-year retention, monthly partitioning, indexed by timestamp and severity.
- Zero-downtime in-memory ring buffer fallback when ClickHouse is unreachable — zero dropped events.

13. Fleet Management & Remote Upgrades

- Remote WPK agent upgrades with cryptographic signature validation.
- Distributed task coordination, health polling, cluster consensus and failover (siem-clusterd).

14. REST API & WebSocket Control Plane (siem-api :8088)

- POST /api/v1/ingest, GET /api/v1/agents, inventory, SCA, parsers, active-response, reports.
- WebSocket /ws/events for live alerts, FIM diffs, and detections to Angular SOC UI.

15. 37-Crate Rust Traceability Matrix (Summary)

Complete modular Cargo workspace: siem-api, siem-engine, siem-parser-gen, siem-vuln-detector, siem-syscollector, siem-agent, siem-agent-linux, siem-syscheckd, siem-execd, siem-integratord, siem-csyslogd, siem-wdb, siem-cdb, siem-authd, siem-crypto, siem-remoted, siem-analysisd, siem-monitord, siem-logtest, siem-maild, siem-reportd, siem-agent-upgrade, siem-task-manager, siem-harvester, siem-rootcheck, siem-clusterd, siem-indexer-connector, siem-event-hub, siem-types, siem-config, siem-metrics, siem-geo, siem-rule-compiler, siem-threat-intel, siem-archive, siem-agent-simulator, provigil-common. All 37 crates compile cleanly with zero warnings.

PART B — NEXT-GENERATION INNOVATIVE FEATURE ROADMAP

The following ten capabilities are designed to differentiate AETHER SIEM from current commercial leaders. These features are largely absent or only partially implemented in platforms as of 2026 and represent the strategic path to top-tier autonomous security operations.

1. Living Detection Genome

Detections are treated as living systems that continuously evolve. AI generates rule variants, tests them against historical data and Red Agent simulations, retains high-performing variants, and retires weak or noisy ones. The detection surface improves automatically over time without constant human rule authoring.

- Health scoring for every detection (TPR, FPR, coverage, age)
- Automated mutation and selection of detection logic
- Integration with Red Agent outcomes for real adversarial pressure

2. Causal Attack Graph + Counterfactual Engine

Moves beyond correlation timelines to true cause-and-effect reasoning. Constructs causal graphs of attacks and answers counterfactual questions such as “If MFA had been enabled, at which step would this attack have failed?” Provides both deeper investigation insight and quantifiable prevention guidance.

- Root-cause identification across multi-stage campaigns
- What-If simulation of missing controls and their impact
- Highest-impact control ranking for risk reduction

3. Continuous Red Agent Swarm

AI-driven red agents continuously attempt realistic attacks against the monitored environment using current techniques. Successful evasions automatically generate new detections and hardening recommendations. The SIEM becomes a perpetual purple-team engine.

- Always-on adversarial simulation aligned with MITRE ATT&CK;
- Automatic detection gap discovery and rule synthesis
- Safe, controlled execution with full audit logging

4. Organizational Security Memory

Long-term episodic and semantic memory of the organization’s security history. Remembers past investigations, analyst decisions, false-positive rationales, and successful response patterns. When similar activity reappears, relevant historical context is surfaced instantly.

- Cross-year recall of similar incidents and resolutions
- Preservation of institutional knowledge beyond individual analysts
- Accelerated triage through historical precedent matching

5. Intent Prediction Engine

Early inference of attacker goals (ransomware staging, data theft, long-term espionage, destructive intent) from weak signals before full impact materializes. Enables preemptive micro-containment and priority elevation.

- Multi-hypothesis intent scoring with confidence levels
- Linkage of early TTPs to probable end objectives
- Triggering of preemptive defensive actions

6. Business-Process Risk Fabric

Dynamic risk scoring that incorporates real business context rather than pure technical severity. The same technical alert receives different priority based on whether it affects a payment system during financial close, a key executive account, or a non-production test server.

- Mapping of assets and identities to business processes
- Time-window and criticality-aware prioritization
- Executive-ready impact narratives

7. Neurosymbolic Decision Engine

Hybrid architecture combining neural pattern recognition with symbolic logic and formal constraints. Every significant AI decision is accompanied by a transparent, machine-checkable reasoning chain suitable for auditors and regulators.

- Explainable decision paths for high-impact actions
- Hard policy and compliance guardrails around autonomous behavior
- Improved trust and regulatory acceptance

8. Analyst Cognitive Load & Trust Calibrator

Monitors analyst workload and interaction patterns to detect fatigue and over-reliance on automation. Dynamically adjusts the level of AI autonomy and the depth of explanations to maintain optimal human-AI collaboration.

- Workload-aware autonomy adjustment
- Detection of automation bias and under-utilization
- Personalized explanation depth based on analyst state

9. Deception-as-Code + Adaptive Honeynet

Automatic deployment and evolution of high-fidelity decoys, honeytokens, and canary resources based on current threat landscape and asset criticality. Attacker interactions feed directly into detection evolution and threat intelligence.

- Context-aware deception asset placement
- Real attacker behavior capture and feedback into Living Genome
- Low false-positive high-signal detection source

10. Federated Privacy-Preserving Threat Swarm

Multiple independent AETHER deployments share only anonymized attack fingerprints and model updates (never raw logs). When one organization observes a novel technique, others rapidly gain protective coverage while fully preserving data sovereignty.

- Differential-privacy and fingerprint-only sharing model
- Rapid collective defense against emerging threats
- Optional participation with full data-sovereignty controls

Implementation Priority for Maximum Impact

Prioritization balances strategic differentiation, technical feasibility within the existing Rust + ClickHouse architecture, and time-to-value.

Rank	Feature	Strategic Value	Difficulty
1	Living Detection Genome	Extremely High	High
2	Causal + Counterfactual Engine	Extremely High	High
3	Continuous Red Agent Swarm	Extremely High	Very High
4	Organizational Security Memory	Very High	Medium-High
5	Intent Prediction Engine	Very High	High
6	Business-Process Risk Fabric	High	Medium
7	Neurosymbolic Decision Engine	High	High

Rank	Feature	Strategic Value	Difficulty
8	Analyst Cognitive Load Calibrator	High	Medium
9	Deception-as-Code	High	Medium-High
10	Federated Threat Swarm	High (network effect)	High

Strategic Conclusion

The production core (Part A) establishes a high-performance, memory-safe, Wazuh-compatible SIEM foundation with a best-in-class autonomous log parser synthesizer. The Next-Generation Innovative Feature Roadmap (Part B) provides the strategic AI-native capability layer required for long-term market leadership. Implementation of the top four features — Living Detection Genome, Causal Counterfactual Engine, Continuous Red Agent Swarm, and Organizational Security Memory — will create compounding defensive advantages that are extremely difficult for competitors to replicate.

This document is the authoritative feature catalog for Provigil AI AETHER SIEM & XDR Platform Version 2.1.